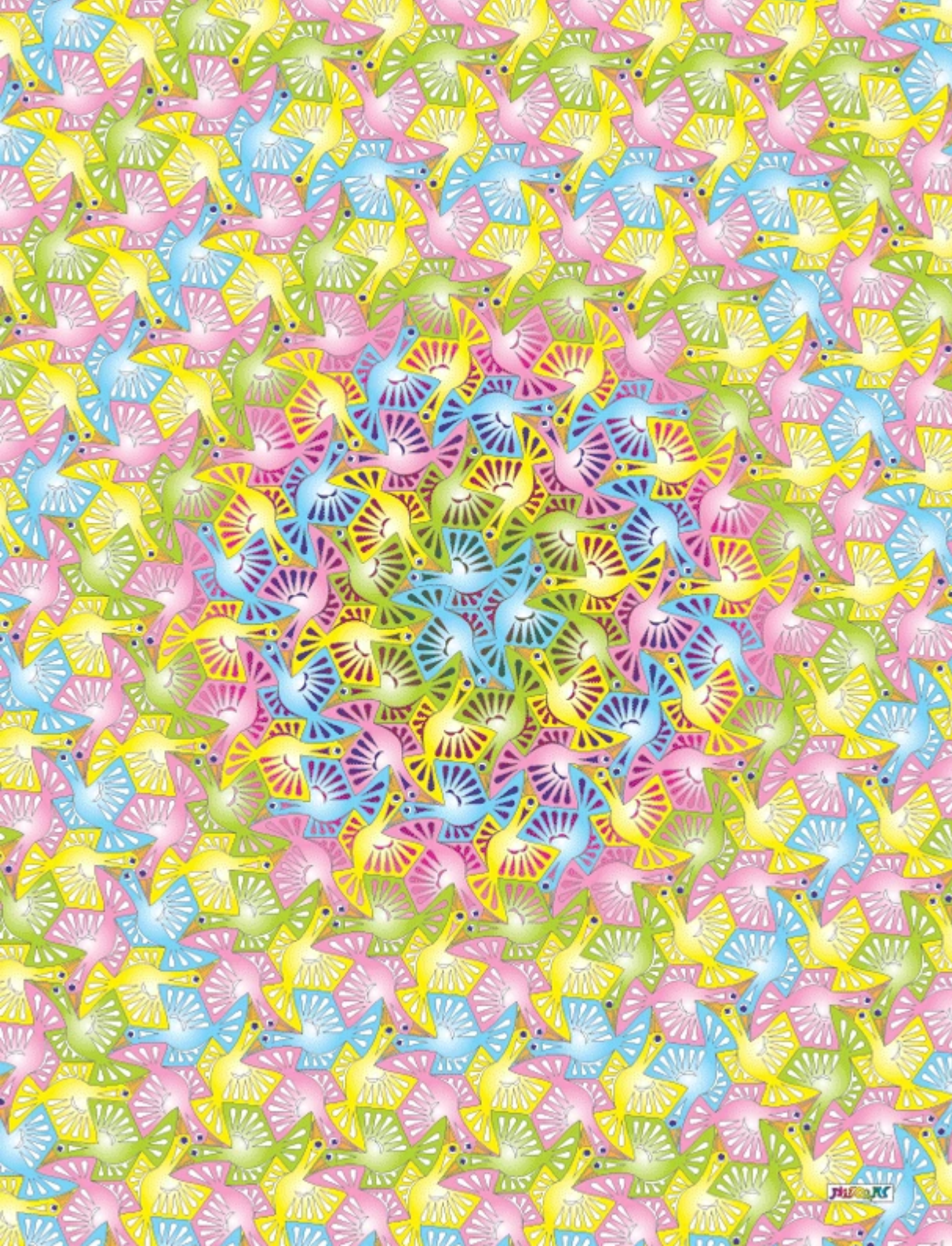




Statistics

Lecture 6

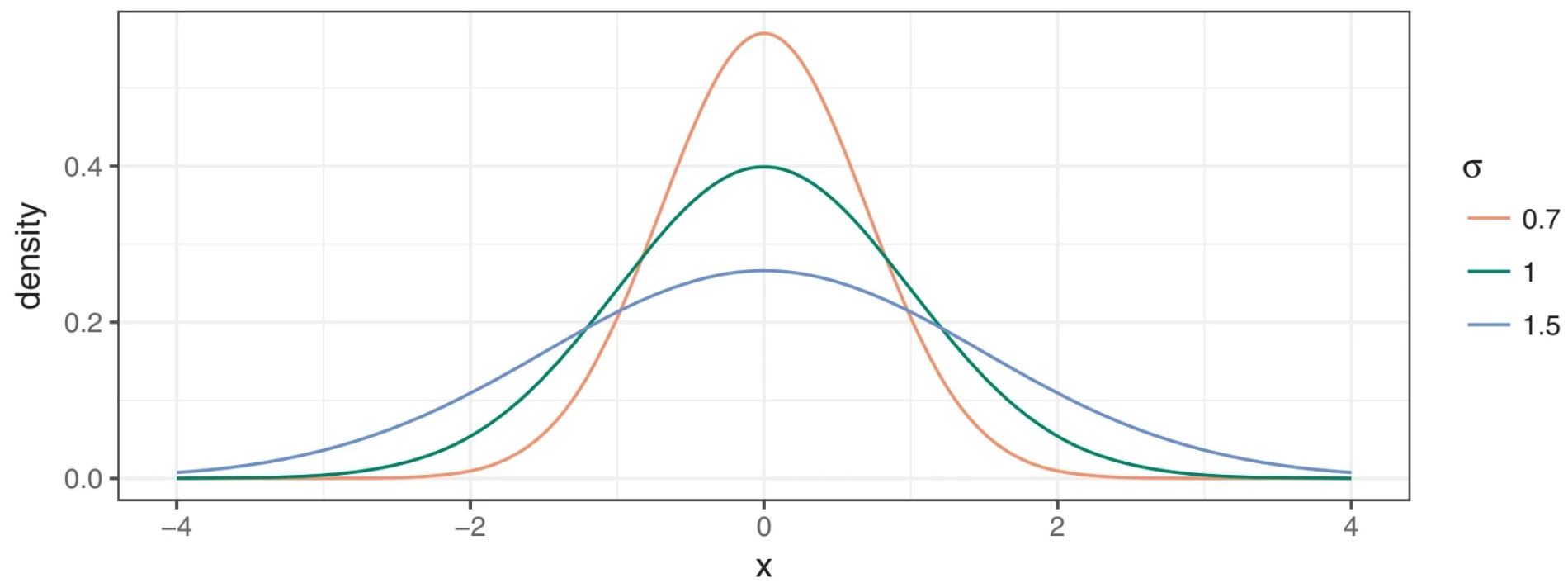
Recall

Distributions derived from normal distributions

The density for normal distribution $\text{Normal}(\mu, \sigma^2)$
is given by $\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.

Its m.g.f is

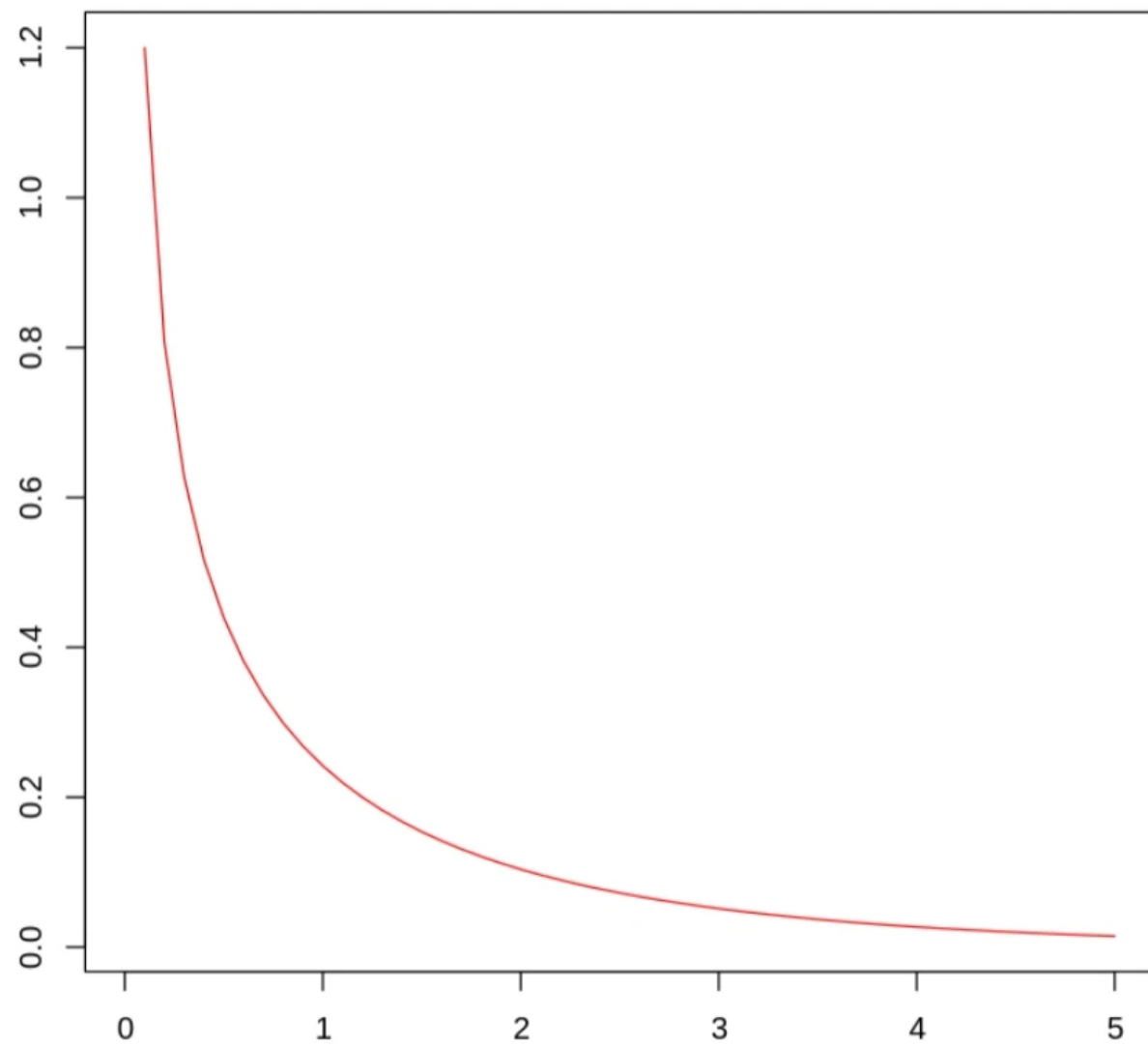
$$M_x(t) = e^{\mu t + \sigma^2 t^2 / 2}$$



χ^2 distribution

If Z is a standard normal distribution, the distribution

$U = Z^2$ is called the χ^2 distribution with 1 degree of freedom.

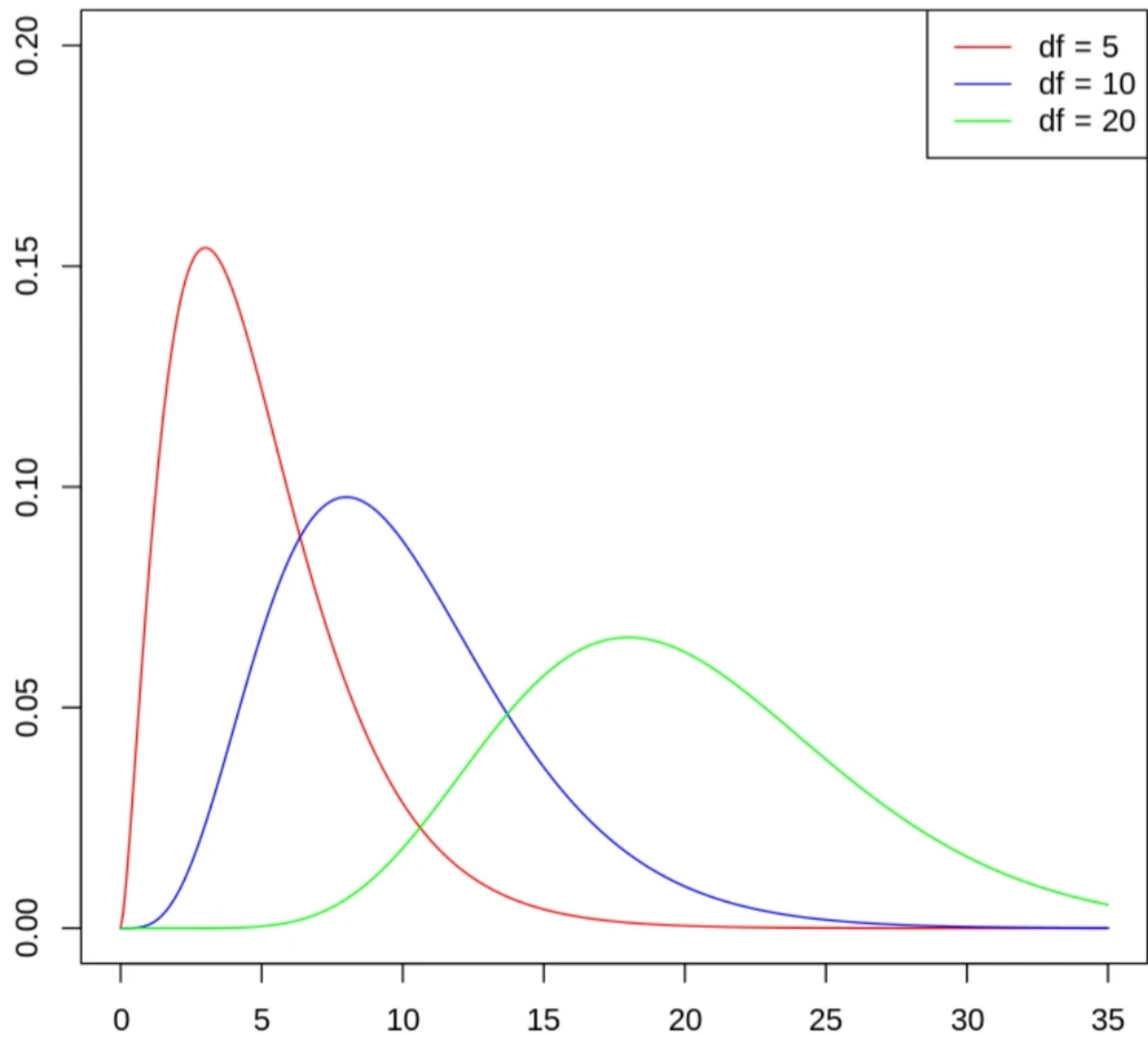


If $U_1, U_2, \dots, U_n$ are independent χ^2 distribution random variables with 1 degree of freedom, the

random variable

$V = U_1 + U_2 + \dots + U_n$ is called the

χ^2 distribution with n degrees of freedom and written as χ_n^2

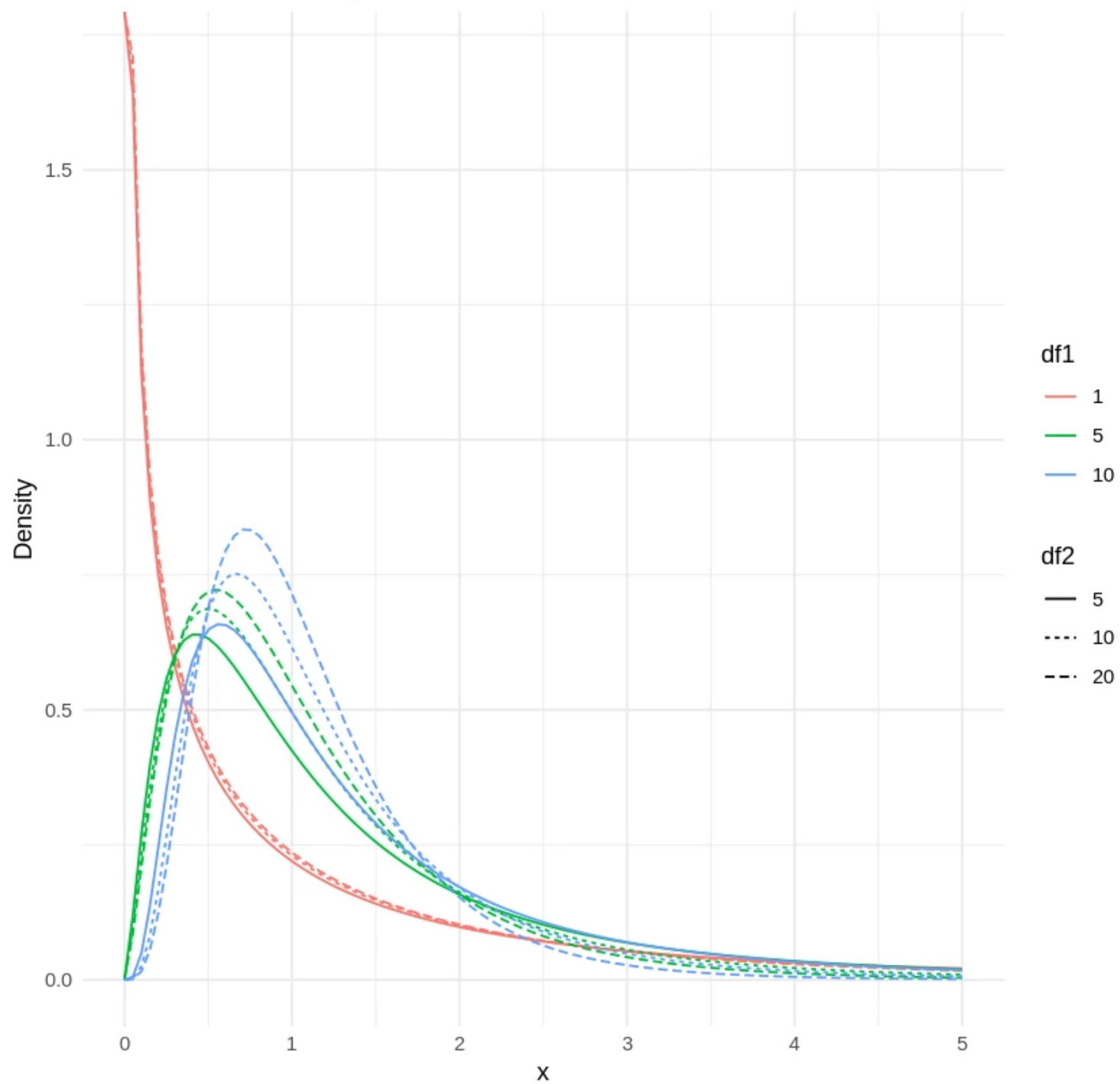


F distribution

Let $Z = U/m / V/n$ where

U and V are independent χ^2 random variables with m and n degrees of freedom respectively. Then Z is said to follow F distribution denoted $F_{m,n}$

F Distribution Density

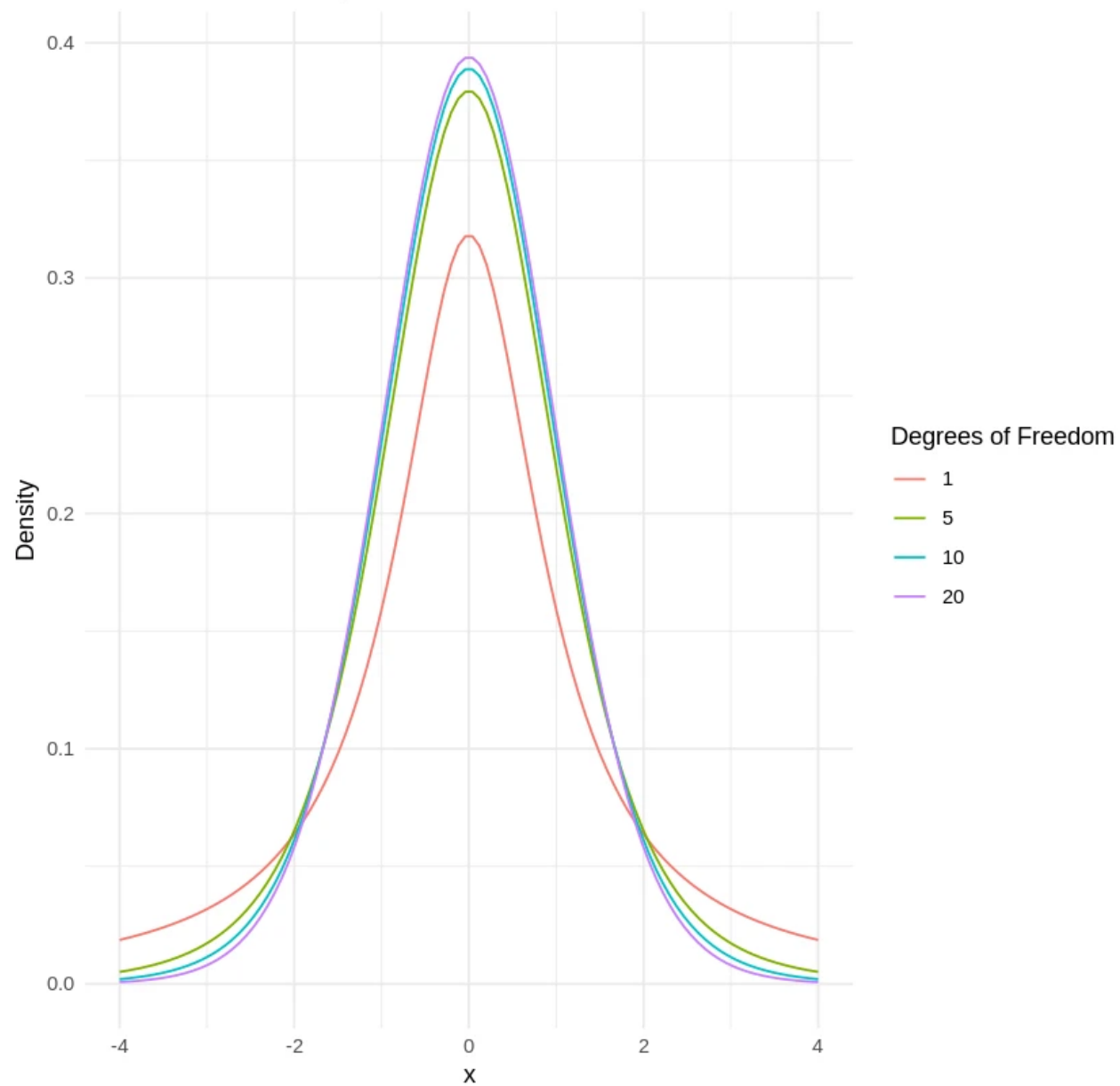


t distributions

Let Z be a Normal $(0,1)$ r.v. and
let U be a χ_n^2 r.v. Then $Y = \frac{Z}{\sqrt{U/n}}$

is called the t -distribution with n
degrees of freedom.

t Distribution Density



Sample mean and

sample variance.

Recall the Sample mean

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

↳ Sample Variance

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

Recall the Sample mean

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

$$E(\bar{X}) = E(X_i) = \mu$$

Sample Variance

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$$

$$E(S^2) = \sigma^2 = \text{var}(X_i)$$

Theorem For $n \geq 2$, let $X_1, X_2, \dots, X_n$ be i.i.d. r.v.s with distribution $X \sim \text{Normal}(\mu, \sigma^2)$. Then

(a) $\bar{X}$ is a normal distribution with mean μ and variance σ^2/n

(b) $\frac{(n-1)S^2}{\sigma^2}$ has the χ^2_{n-1} distribution

(c) $\bar{X}$ and S^2 are independent.

Proof

(a) This follows from the properties of the m.g.f.

$$\text{m.g.f.} \left(\frac{X_1 + X_2}{2} \right) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

which is the m.g.f. for

$N\left(\mu, \frac{\sigma^2}{2}\right)$. The rest follows from induction.

(c) it is enough to prove that $\bar{X}$ and the vector $(x_1 - \bar{X}, x_2 - \bar{X}, \dots, x_n - \bar{X})$ are independent.

This will be shown by demonstrating that the m.g.f $M(s, t_1, \dots, t_n)$
 $= E[\exp(s\bar{X} + t_1(x_1 - \bar{X}) + \dots + t_n(x_n - \bar{X}))]$
factors into a product.

Step 1 $\sum_{i=1}^n t_i (x_i - \bar{X}) = \sum_{i=1}^n t_i x_i - n\bar{X}\bar{t}$

(c) Step 1 $\sum_{i=1}^n t_i (x_i - \bar{x}) = \sum_{i=1}^n t_i x_i - n \bar{x} \bar{t}$

then $s\bar{x} + \sum_{i=1}^n t_i (x_i - \bar{x})$

$$= \sum_{i=1}^n \left[\frac{s}{n} + (t_i - \bar{t}) \right] x_i$$

$$= \sum_{i=1}^n a_i x_i \quad \text{where}$$

$$a_i = \frac{s}{n} + t_i - \bar{t}$$

Step 2

$$\sum_{i=1}^n a_i = s$$

and

$$\sum_{i=1}^n a_i^2 = \frac{s^2}{n} + \sum_{i=1}^n (t_i - \bar{t})$$

(the cross term
vanishes!)

Step 3

$$M(s, t_1, t_2, \dots, t_n)$$

$$= M\left(\exp\left(\sum_{i=1}^n a_i x_i\right)\right)$$

$$= M\left(\prod_{i=1}^n \exp a_i x_i\right)$$

$$= \prod_{i=1}^n M_{x_i}(a_i)$$

$$= \prod_{i=1}^n \exp\left(\mu a_i + \frac{\sigma^2}{2} a_i^2\right)$$

$$= \exp\left(\mu \sum_{i=1}^n a_i + \frac{\sigma^2}{2} \sum_{i=1}^n a_i^2\right)$$

Step 3

$$M(s, t_1, t_2, \dots, t_n)$$

$$= \exp\left(\mu \sum_{i=1}^n a_i + \frac{\sigma^2}{2} \sum_{i=1}^n a_i^2\right)$$

$$= \exp\left(\mu s + \frac{\sigma^2}{2} \frac{s^2}{n} + \frac{\sigma^2}{2} \sum_{i=1}^n (t_i - \bar{t})^2\right)$$

$$= \exp\left(\mu s + \frac{\sigma^2}{2} \frac{s^2}{n}\right) \exp\left(\frac{\sigma^2}{2} \sum_{i=1}^n (t_i - \bar{t})^2\right)$$

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m.g.f of $\bar{X}$

joint m.g.f of
 $(X_1 - \bar{X}, X_2 - \bar{X}, \dots, X_n - \bar{X})$

(b) We want to show that $\frac{(n-1)S^2}{\sigma^2}$ has
distribution χ_{n-1}^2

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$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi_n^2$$

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Also,

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{\sigma^2} \sum [(x_i - \bar{x})^2 + (\bar{x} - \mu)^2]$$

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$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n \left(\frac{x_i - \mu}{\sigma} \right)^2 \sim \chi_n^2$$

Also,
$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{\sigma^2} \sum [(x_i - \bar{x})^2 + (\bar{x} - \mu)^2]$$

Expanding the square:

$$= \frac{1}{\sigma^2} \sum (x_i - \bar{x})^2 + \frac{(\bar{x} - \mu)^2}{\sigma^2/n}$$

(b) We want to show that $\frac{(n-1)S^2}{\sigma^2}$ has
distribution χ_{n-1}^2

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{\sigma^2} \sum (x_i - \bar{x})^2 + \frac{(\bar{x} - \mu)^2}{\sigma^2/n}$$

$$W = U + V$$

(b) We want to show that $\frac{(n-1)S^2}{\sigma^2}$ has distribution χ_{n-1}^2

$$\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{\sigma^2} \sum (x_i - \bar{x})^2 + \frac{(\bar{x} - \mu)^2}{\sigma^2/n}$$

$$W = U + V$$

By independence of $\bar{X}$ and S ,

$$\text{mgf}(W) = \text{mgf}(U) \cdot \text{mgf}(V)$$

$$\therefore \text{mgf}(U) = \frac{\text{mgf}(W)}{\text{mgf}(V)} = \frac{(1-2t)^{-n/2}}{(1-2t)^{-1/2}}$$



Corollary Let $\bar{X}$ and S^2 be the sample mean & sample variance $\Rightarrow$ then

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

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$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

Proof

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)}{\sqrt{\frac{S^2}{\sigma^2}}} = \frac{\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right)}{\sqrt{\frac{1}{n-1} \frac{(n-1)S^2}{\sigma^2}}}$$

Sampling

Examples

- ① Government Census, health surveys
- ② Quality Control
- ③ Audits of Financial Records of Large Companies
- ④ Agriculture
- ⑤ Traffic

Why sampling?

- ① Random sampling prevents against bias.
- ② Working with smaller samples is faster.
- ③ Working with large samples can introduce errors at the level of cleaning up the data and would require training a large number of people.
- ④ The knowledge of sampling techniques lets us make error estimates and these error estimates allow us to decide on the size of samples required.

Population Parameters.

These are the quantities such as population mean that we want to estimate from the sample.

We will assume that pop. has size N and that each member of the population is assigned a numerical value such as age, weight, height, marital status etc. These num. are denoted by $x_1, x_2, \dots, x_N$.

An example. (Herkson, 1976) from John Rice's book.

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Population - $N = 393$ short stay hospitals.

- Let x_i denote the number of patients discharged from the i^{th} hospital during

January 1968.

- A histogram was constructed from the data.

Bins of size 200 were constructed

0-200, 201-400, ..., 2801-3000 - Horizontal bars of height equal to the no. of hospitals that discharged patients in that range.

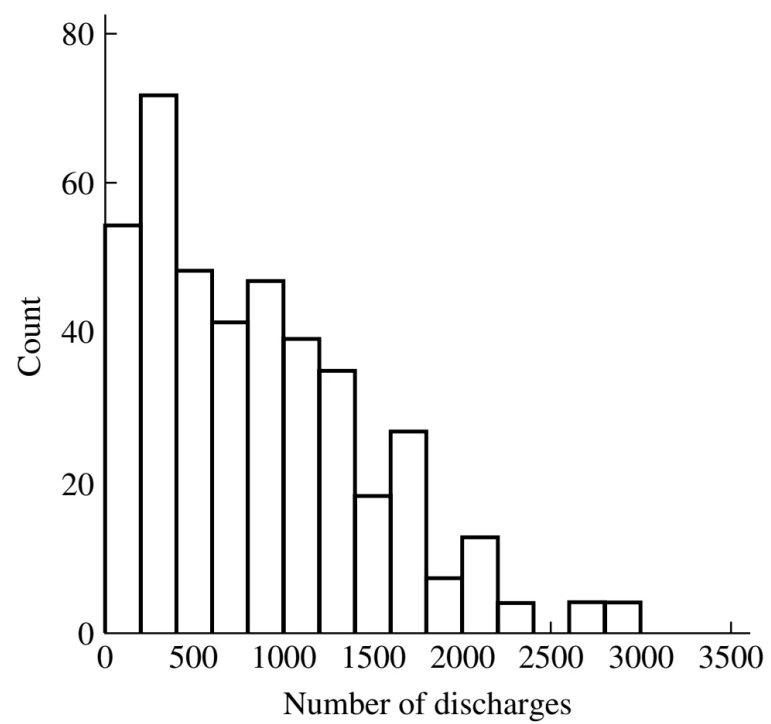


FIGURE 7.1 Histogram of the numbers of patients discharged during January 1968 from 393 short-stay hospitals.

Population Parameters

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

$$T = \sum_{i=1}^N x_i = N\mu$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 = \frac{\sum_{i=1}^N x_i^2}{N} - \mu^2$$

Population Parameters

pop. mean

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

pop. total

$$T = \sum_{i=1}^N x_i = N\mu$$

pop. variance.

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 = \frac{\sum_{i=1}^N x_i^2}{N} - \mu^2$$

Simple Random Sampling

Each sample of size n has the same probability of occurrence.

Each of the ${}^N C_n$ possible samples of size n taken without replacement are equally likely.

Each number of the population can appear at most once.

The particular members would be chosen using a random number generator.

The sample members are denoted by $X_1, X_2, \dots, X_n$ and each of them is a random variable.

Sample mean

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

is a random variable.

So is the sample total

$$T = \sum_{i=1}^n X_i = N \bar{X}$$

The distribution of $\bar{X}$ and T are called sampling distributions

Going back to the example of 393 hospitals.

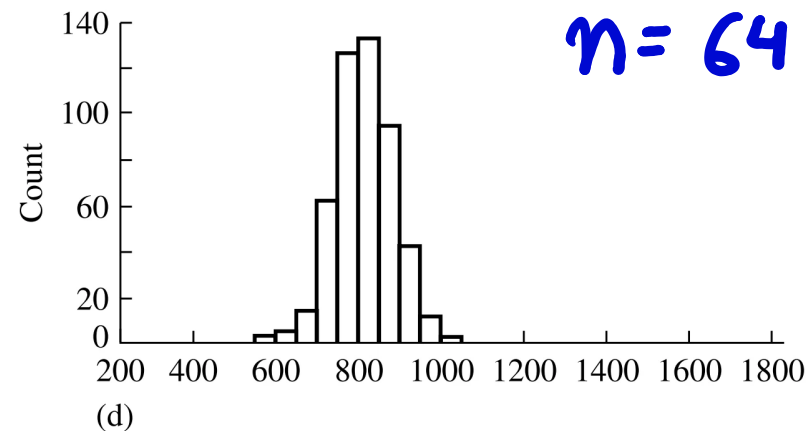
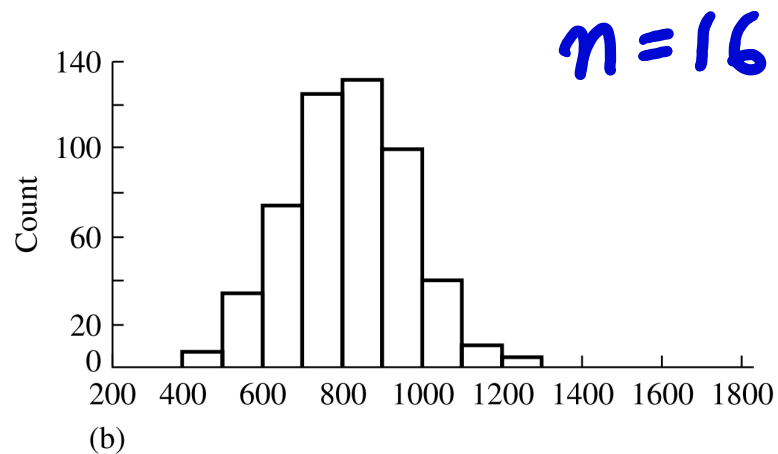
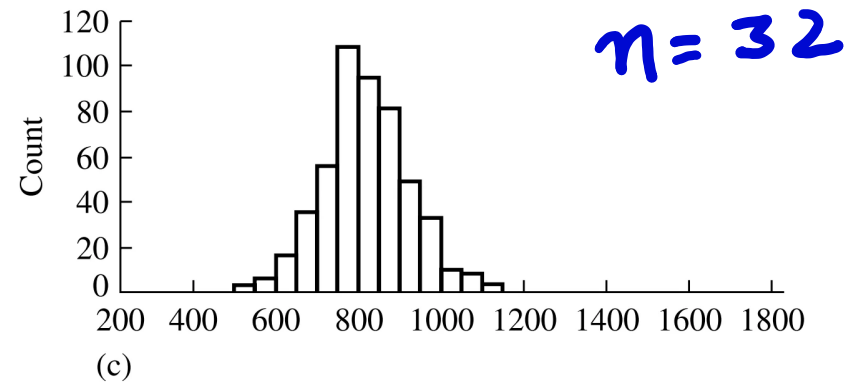
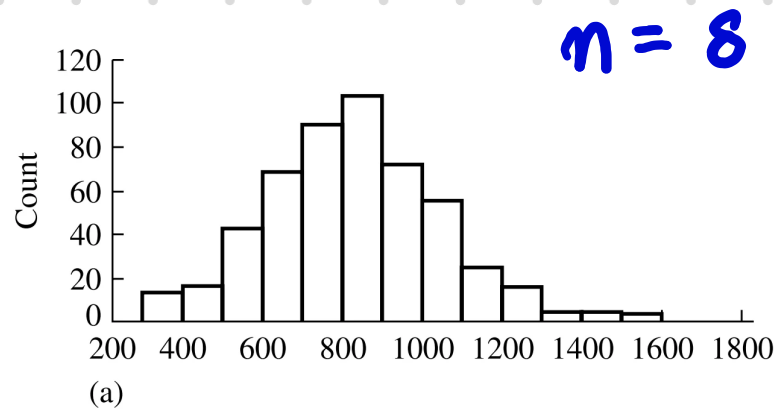
To find the sampling distribution of the sample mean of size 16, one would look at ${}^{393}C_{16}$ samples and compute the mean of each one of them.

Going back to the example of 393 hospitals.

To find the sampling distribution of the sample mean of size 16, one would look at ${}^{393}C_{16}$ samples and compute the mean of each one of them.

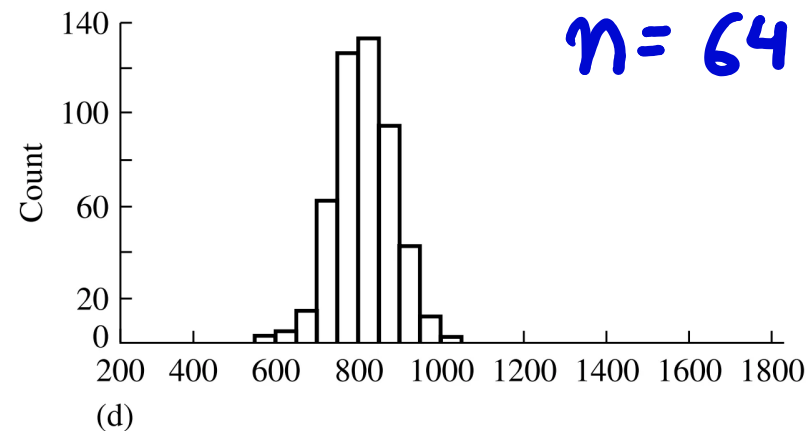
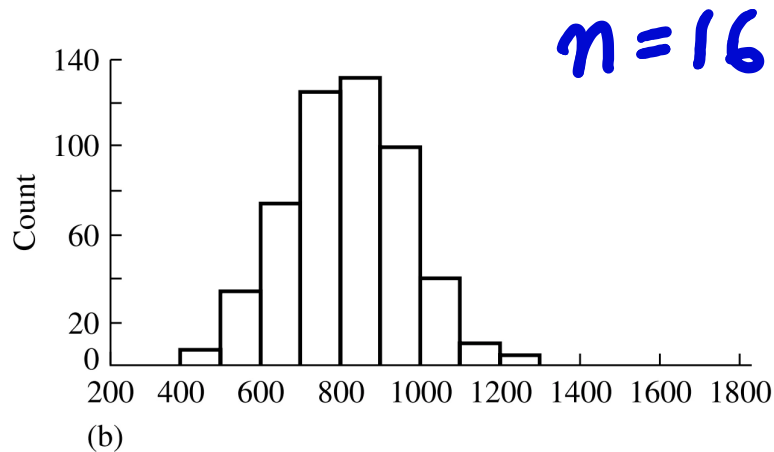
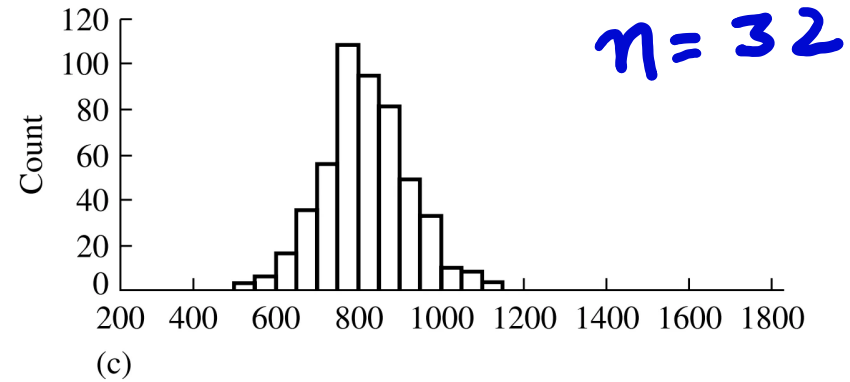
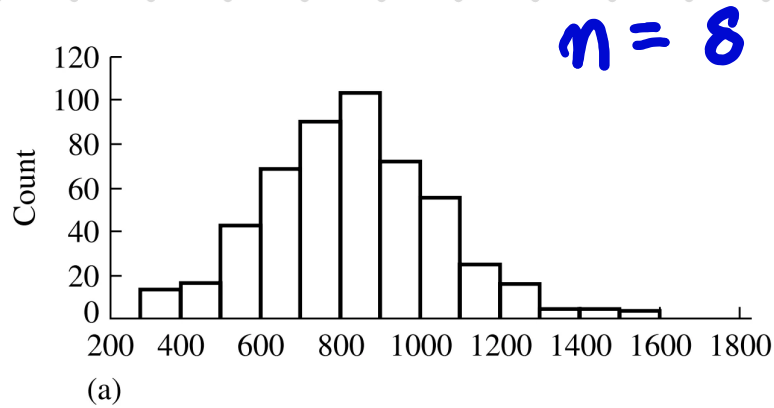
← This is not practical.

One can simulate what the sampling distribution might look like by taking many samples of size n and then drawing a histogram.



500 replications for every sample size.

- ① All histograms are centered around the pop. mean
- ② less spreading for larger n .
- ③ more symmetric for larger n .



500 replications for every sample size.

We can justify these observations mathematically.

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Denote the distinct values assumed the population numbers by $\xi_1, \xi_2, \dots, \xi_m$ and denote the number of population numbers that have that value ξ_j by n_j for $j=1, 2, \dots, m$. Then X_i is a discrete random variable with p.m.f.

$$P(X_i = \xi_j) = \frac{n_j}{N}, \quad E[X_i] = \mu, \\ \text{Var}[X_i] = \sigma^2$$

Proof ○ Since each member of the population is equally likely to be in the sample.

$$\therefore P(X_i = \xi_j) = \frac{n_j}{N} \quad \text{for } j=1, 2, \dots, m.$$

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$$\begin{aligned} \text{◎ } E[X_i] &= \sum_{j=1}^m \xi_j P(X_i = \xi_j) \\ &= \sum_{j=1}^m \xi_j \times \frac{n_j}{N} = \mu. \end{aligned}$$

Proof ① Since each member of the population is equally likely to be in the sample.

$$\therefore P(X_i = \xi_j) = \frac{n_j}{N} \text{ for } j=1, 2, \dots, m.$$

$$\begin{aligned} \textcircled{2} \quad E[X_i] &= \sum_{j=1}^m \xi_j P(X_i = \xi_j) \\ &= \sum_{j=1}^m \xi_j \times \frac{n_j}{N} = \mu. \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad \text{Var}[X_i] &= E[X_i^2] - \mu^2 = \frac{1}{N} \sum \xi_j^2 n_j - \mu^2 \\ &= \frac{\sum_{k=1}^N x_k^2}{N} - \mu^2 = \sigma^2. \end{aligned}$$

Theorem.

With simple random
sampling, $E[\bar{X}] = \mu$.

Theorem. With simple random sampling, $E[\bar{x}] = \mu$.

Proof

$$E[\bar{x}] = \frac{1}{n} \sum_{i=1}^n E[x_i] \\ = \mu.$$

Theorem.

With simple random
sampling, $E[T] = \tau$.

Theorem. With simple random sampling, $E[T] = T$.

Proof

$$\begin{aligned} E[T] &= E[N\bar{x}] \\ &= N E[\bar{x}] \\ &= N \cdot \mu \\ &= T \rightarrow \text{total} \end{aligned}$$

For simple random sampling.

Theorem: $\text{Cov}(x_i, x_j) = \frac{-\sigma^2}{N-1}$ if $i \neq j$

For simple random sampling.

Theorem: $\text{Cov}(X_i, X_j) = \frac{-\sigma^2}{N-1}$ if $i \neq j$

Proof

$$\text{Cov}(X_i, X_j) = E(X_i X_j) - E(X_i)E(X_j)$$


$$\begin{aligned} E(X_i X_j) &= \sum_{k=1}^m \sum_{l=1}^m z_k z_l P(X_i = z_k \text{ and } X_j = z_l) \\ &= \sum_{k=1}^m z_k P(X_i = z_k) \sum_{l=1}^m z_l P(X_j = z_l | X_i = z_k) \end{aligned}$$

For simple random sampling.

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$$P(X_j = \xi_l \mid X_i = \xi_k) = \begin{cases} & \text{if } k \neq l \\ & \text{if } k = l \end{cases}$$

$$P(X_j = \xi_l \mid X_i = \xi_k) = \begin{cases} n_l / (N-1) & \text{if } k \neq l \\ (n_l - 1) / (N-1) & \text{if } k = l \end{cases}$$

$$P(X_j = \xi_l \mid X_i = \xi_k) = \begin{cases} n_l / (N-1) & \text{if } k \neq l \\ (n_l - 1) / (N-1) & \text{if } k = l \end{cases}$$

$$\therefore \sum_{l=1}^m \xi_l P(X_j = \xi_l \mid X_i = \xi_k)$$

$$= \sum_{l \neq k} \xi_l \frac{n_l}{N-1} + \xi_k \frac{n_k - 1}{N-1}$$

$$= \sum_{l=1}^m \xi_l \frac{n_l}{N-1} - \frac{\xi_k}{N-1}$$

For simple random sampling.

Theorem: $\text{Cov}(X_i, X_j) = \frac{-\sigma^2}{N-1}$ if $i \neq j$

Proof

$$\text{Cov}(X_i, X_j) = E(X_i X_j) - E(X_i)E(X_j)$$


$$\begin{aligned} E(X_i X_j) &= \sum_{k=1}^m \sum_{l=1}^m z_k z_l P(X_i = z_k \text{ and } X_j = z_l) \\ &= \sum_{k=1}^m z_k P(X_i = z_k) \underbrace{\sum_{l=1}^m z_l P(X_j = z_l | X_i = z_k)}_{?} \end{aligned}$$

$$E(X_i X_j) = \sum_{k=1}^m \sum_{l=1}^m z_k z_l P(X_i = z_k \text{ and } X_j = z_l)$$

$$= \sum_{k=1}^m z_k P(X_i = z_k) \sum_{l=1}^m z_l P(X_j = z_l | X_i = z_k)$$

$$= \sum_{k=1}^m z_k \frac{n_k}{N} \left(\sum_{l=1}^m z_l \frac{n_l}{N-1} - \frac{z_k}{N-1} \right)$$

$$= \frac{T^2}{N(N-1)} - \frac{\sum_{k=1}^m z_k^2 n_k}{N(N-1)}$$

$$= \frac{N\mu^2}{N-1} - \left(\frac{\mu^2 + \sigma^2}{N-1} \right) = \mu^2 - \frac{\sigma^2}{N-1}$$

For simple random sampling.

Theorem: $\text{Cov}(x_i, x_j) = \frac{-\sigma^2}{N-1}$ if $i \neq j$

Proof $\text{Cov}(x_i, x_j) = E(x_i x_j) - E(x_i)E(x_j)$

A curved arrow points from $E(x_i x_j)$ in the equation above to $\mu^2 - \frac{\sigma^2}{N-1}$ in the equation below. A straight arrow points from $E(x_i)E(x_j)$ in the equation above to μ^2 in the equation below.

$$E(x_i x_j) = \mu^2 - \frac{\sigma^2}{N-1}$$
$$\mu^2$$

Therefore, $\text{cov}(x_i, x_j) = \frac{-\sigma^2}{N-1}$ for $i \neq j$

Theorem

With simple random sampling

$$\text{Var}(\bar{x}) = \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right)$$

Theorem With simple random sampling

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right)$$

Proof $\text{Var}(\bar{X}) = \text{Cov} \left(\sum_{i=1}^n X_i, \sum_{j=1}^n X_j \right)$

$$= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) + \underbrace{\sum_{i=1}^n \sum_{i \neq j} \text{Cov}(X_i, X_j)}_{n^2}$$

$$= \frac{\sigma^2}{n} - \frac{n(n-1)\sigma^2}{n^2(N-1)} = \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right)$$

Theorem

With simple random sampling

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right)$$

Proof

$$\text{Var}(\bar{X}) = \text{Cov} \left(\sum_{i=1}^n X_i, \sum_{j=1}^n X_j \right)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j)$$

$$= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) + \underbrace{\sum_{i=1}^n \sum_{i \neq j} \text{Cov}(X_i, X_j)}_{n^2}$$

$$= \frac{\sigma^2}{n} - \frac{n(n-1)\sigma^2}{n^2(N-1)} = \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right)$$

Notice the difference
between sampling with replacement

$$\frac{\sigma^2}{n}$$

and sampling without replacement

$$\frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right)$$

This is a correction factor.
Generally the factor $\frac{n}{N}$ is called the sampling fraction.

Notice the difference
between sampling with replacement

$$\frac{\sigma^2}{n}$$

and sampling without replacement

$$\frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right)$$

Generally the population size N
is very large so that $\sigma_{\bar{x}}^2 \approx \frac{\sigma^2}{n}$

This is a correction factor.

Generally the factor $\frac{n}{N}$ is called the sampling fraction.

Theorem

With random sampling.

$$\text{Var}(T) = N^2 \left(\frac{\sigma^2}{n} \right) \left(\frac{N-n}{N-1} \right)$$

Theorem With random sampling.

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Proof

$$T = N\bar{X}$$

$$\begin{aligned} \therefore \text{Var}(T) &= N^2 \text{Var}(\bar{X}) \\ &= N^2 \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) \end{aligned}$$

In the case of Sampling with replacement (i.i.d. case), we saw that

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n} \text{ is unbiased}$$

which is to say that its expected value is different from the population variance

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is different from the population

Variance

The unbiased estimator is $\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$.

→ This is also true for simple random sampling.

Theorem With random sampling

$$E[\hat{\sigma}^2] = \sigma^2 \left(\frac{n-1}{n} \right) \frac{N}{N-1}$$

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Proof

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2$$

$$\begin{aligned} \therefore E(\hat{\sigma}^2) &= \frac{1}{n} \sum_{i=1}^n E(x_i^2) - E(\bar{x}^2) \\ &= \frac{1}{n} \sum_{i=1}^n \left(\text{Var}(x_i) + E(x_i)^2 \right) - E(\bar{x}^2) \end{aligned}$$

Theorem With random sampling

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$$= \sigma^2 + \mu^2 - \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right) - \mu^2$$

$$= \sigma^2 \left(1 - \frac{1}{n} + \frac{n-1}{n(N-1)} \right)$$

$$= \sigma^2 \frac{n(N-1) - N + n}{n(N-1)}$$

$$= \frac{\sigma^2 N(n-1)}{n(N-1)}$$



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$$= \sigma^2 \frac{N(n-1)}{n(N-1)}$$

~ This factor is less than 1.
 Therefore this estimator underestimates the population variance.



Therefore, in the case of simple random sampling, an unbiased estimator of σ^2 is

$$\frac{n(N-1)}{N(n-1)} \cdot \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$= \frac{1}{n-1} \cdot \left(1 - \frac{1}{N}\right) \sum_{i=1}^n (x_i - \bar{x})^2$$

Also recall that

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right)$$

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$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right)$$

Therefore an unbiased estimate for $\text{Var}(\bar{X})$ is found from the

estimator

$$S_{\bar{X}}^2 \left\{ \begin{aligned} &= \frac{N-1}{N} \cdot \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1} \cdot \frac{N-n}{N-1} \cdot \frac{1}{n} \\ &= \frac{s^2}{n} \cdot \frac{N-n}{N} \end{aligned} \right.$$